Lecture: 4

Pointer and structure

Revision

Que:1 Why does scanf require &, to scan primitive data types?

Ans:

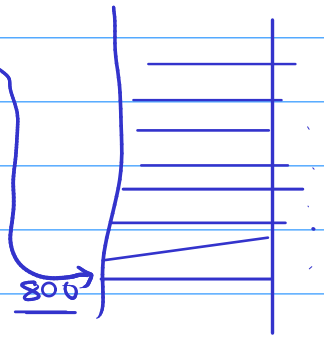
```
void make_it_5(int *ptr) {
    *ptr=5;
}
int main() {
    int x;
    make_it_5(&x);
    printf("%d", x);
}
```

```
scanf( ↓ ) {
    ____
}
int main() {
    int x;
    scanf("%d", &x);
}
```

Que:2 Why & is not required to scan string?

Ans: 2

```
char name[10];
scanf("%s", name);
```



Que:3: find occurences of a string in other string using strstr.

```
char input[ ] = "DDDUDDDU", to_search[ ] = "DDU";
//Strstr( input , to_search);
char *ptr = input;
int cnt = 0;
while(1) {
    ptr = strstr( ptr , to_search);
    if( ptr != NULL) {
        cnt++;
        ptr++;
    }
    else
        break;
}
printf("%d", cnt );
```

AAAAA

AAA

Que:4 what is the difference between

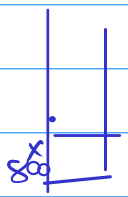
```
int sum(int arr[]){
    int total=0;
    for(int i=0; i<5; i++)
        total += arr[i];
    return total;
}
```

```
int sum2(int *arr){
    int total=0;
    for(int i=0; i<5; i++)
        total += arr[i];
    return total;
}
```

No difference

```
int main(){
    int arr[5]={1,2,30,4,5};
    int total = sum(arr);
    printf("%d", total);
    int total2= sum2(arr);
    printf("%d", total2);
    return 0;
}
```

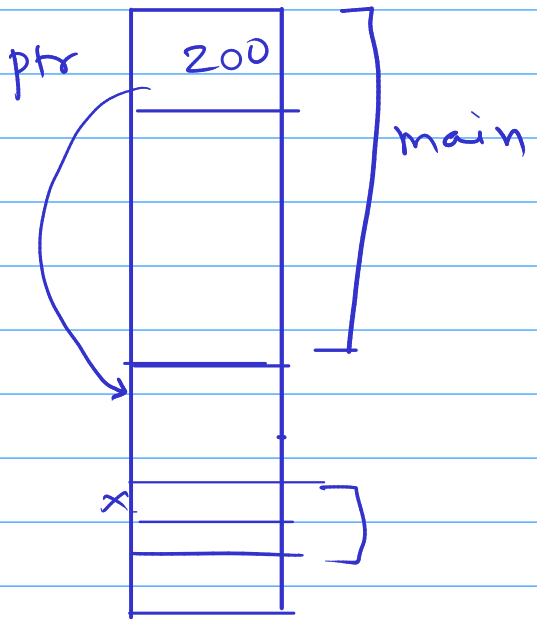
```
int x=7;
sum(&x);
```



→ prefer [] when array is passed. otherwise prefer *.

Que:5 Is following code valid? If no, why?

```
int *fun(){
    int x=7;
    return &x;
}
int main(){
    int *ptr;
    ptr = fun();
    printf("%d", *ptr);
}
```



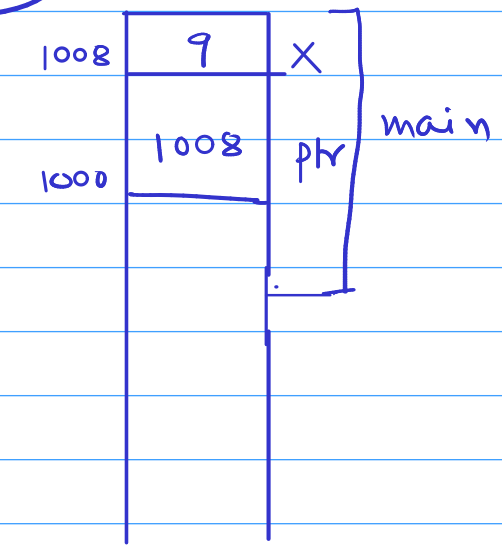
NOTE: Never return address of auto variable

→ Refrain from returning address of static local variable, even when it is allowed.

Que:6 Is following code valid? If no, why?

Yes

```
int *fun(int *ptr){
    *ptr=9;
    return ptr;
}
int main(){
    int x=7, *ptr=&x;
    ptr = fun(&x);
    printf("%d", *ptr);
}
```

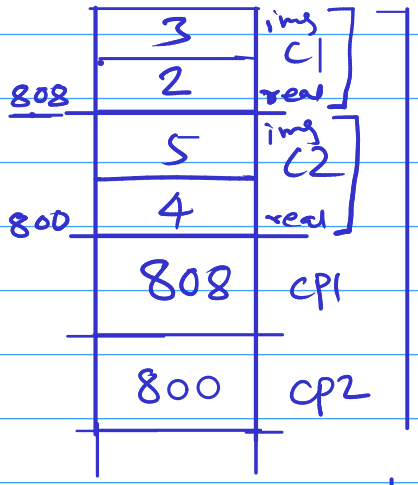


O/P 9

⇒ pointer to a structure variable

```
typedef
struct Cmplx {
    int real;
    int img;
}Complex;

int main() {
    Complex c1 = {2,3}, *cp2;
    struct Cmplx *cp1, c2 = {4,5};
    cp1 = &c1;
    cp2 = &c2;
X   cp1 = &(c1.real);
✓   int *ptr = &(c1.real);
    H.W   &c1.img
```



same

c1 & c2 are of same type
cp1 & cp2 are of same type.

cp1 & cp2 are pointers to Complex type.

```
int x = (*cp1).real;
int x = cp1->real;
```

Same

arrow operator → 1. deference operand on left side
2. Apply dot operator on deferenced variable

```
int main() {
    typedef
    struct {
        int id;
        char name[20];
    }student;
    student s1 = {2, "Rohit"}, *sp = &s1;
X   int id = sp.id;
    int id = sp -> id; // id will get value 2.

X   char ch = sp -> name;
    char *ch = sp -> name;
    char ch1 = sp -> name[2]; // ch will get 'h'.
```